

AI Engineering Summer Training

Daily Task Submission

Topic: What is vibecoding (Week 2 - Day 1)

Submitted by: [Molham waleed]

Student ID: [2338908]

Supervisor: Eng. Abdullah Barghash

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Software Specification: AI-Powered Security+ Quiz Web App

1. The Problem It Solves Preparing for rigorous IT certifications like Security+ requires constant active recall. Traditional question banks eventually become predictable as users memorize the static answers. This simple Web App solves the problem by providing a distraction-free testing environment that leverages AI to generate an infinite variety of unique questions on demand, ensuring genuine comprehension rather than mere memorization.

2. Key Features

- **Customizable Sessions:** Users can select the exact number of questions they want to tackle before the quiz begins.
- **AI-Generated Question Engine:** Dynamically generates fresh, unpredictable multiple-choice questions in real-time using an AI model, replacing static and predictable question banks.
- **In-Depth AI Explanations:** After submitting an answer, the app provides a detailed technical breakdown explaining why the correct choice is right and why the other options are incorrect, reinforcing the learning process.
- **Targeted Topic Focus:** Users can optionally specify a specific Security+ domain (e.g., Cryptography, Network Attacks, Identity Management) to tailor their generated quiz.

3. Constraints (What it should NOT do)

- **No Complex Frontend Frameworks:** To remain feasible for a one-week build, the UI will be strictly built using a Python-only framework (Gradio). It will not use complex web stacks like React or Angular.
- **No User Authentication or Cloud Profiles:** The tool acts as a quick-launch utility and will not require user logins, passwords, or profile creation.
- **No Persistent Databases:** All generated questions and scores are kept temporarily in the session memory and clear when the app restarts, bypassing the need for complex database management.